

Md Samshad Rahman (MACS | M.Sc. Stats | B.Sc. CS)

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Professional summary:

Solutions Engineer (MACS | M.Sc. Stats) with a Dual Master's in Applied Statistics and Computer Science. Expert in modernizing legacy reporting systems into automated, single-source-of-truth data architectures. Proven track record of leading data governance initiatives, designing executive dashboards (Tableau, Dash), and translating complex analytics into evidence-based policy insights for stakeholders. Specializes in cost-effective cloud analytics and automating the full data lifecycle to drive operational excellence.

Relevant Projects & Experiences:

Dalhousie University, Halifax, Nova Scotia

AI-Powered Mindful Eating App (Persuasive Computing Lab)

[January 2025 – April 2025]

- Designed and deployed a production-grade AI inference pipeline by fine-tuning Llama 3.2 (3B), optimizing model parameters and memory constraints to maximize performance, accuracy, and cost-efficiency on T4 GPUs.
- Proactively addressed AI system reliability and hallucinations by establishing a risk-mitigation framework, utilizing a curated dataset of 1,500 expert-verified inputs to validate model outputs for enterprise safety.
- Architected the end-to-end system linking a React Native mobile interface to a FastAPI microservice. I designed the backend to handle multi-modal inputs (text and photos) in real-time, allowing users to log meals and receive instant personality-based feedback with low latency.
- Implemented psychographic personalization by integrating the Symanto NLP API to detect “Big Five” personality traits from user text. This enabled the system to programmatically modulate its tone and advice strategy for each user, solving the problem of generic, non-specific guidance.

Awarded Public Tenders Analysis Dashboard (Visual Analytics)

[September 2024 – December 2024]

- Fixed a broken NS government reporting process by writing Python scripts to ingest, clean, and standardize 10 years of raw Nova Scotia Open Data tender records.
- Engineered validation logic that corrected formatting errors and reduced dataset “noise” by ~35%, giving analysts a reliable single source of truth for the first time.
- Built a custom intelligence tool using Dash/Plotly that clustered unstructured contract descriptions into coherent themes, cutting research time by ~40%.
- Established clear data dictionaries and documentation to ensure future maintainability of the pipeline.

Systems Analyst (Data Integrity & Billing), Munshi Engineers, Bangladesh

[February 2021 – April 2023]

- Managed critical billing infrastructure for national power sectors (DPDC, BPDB), ensuring the integrity of high-volume financial data used for commercial support and trend analysis.
- Engineered Python/SQL health checks to monitor consumption trends and flag statistical anomalies in billing data before invoice generation.
- Executed rigorous export/import validation workflows to maintain a 100% accuracy rate on sensitive utility records.
- Transformed raw operational logs into actionable monthly reports using Excel and SQL, enabling management to spot revenue leakage.

Python Developer, Smartbytes Ltd. Bangladesh

[February 2022 – September 2022]

Resume Matching with Job Descriptions (German Job Market)

- Eliminated manual screening bottlenecks by engineering an automated text-processing engine (NLP) that parsed unstructured resumes into structured fields (PostgreSQL).
- Configured the API to auto-generate documentation (Swagger), reducing integration time for the frontend team.
- Benchmarked multiple classification models to select LinearSVC, boosting data matching accuracy to 92%.

Research Assistant, Cognitive & Behavioral Data Science Lab, UI University, Bangladesh [October 2021 – March 2022]

Eating Preference Analysis Based on Human Personality Traits – Big5 (Data Science)

- Engineered a resilient ingestion pipeline for social data (Twitter/Foursquare), implementing strict rate-limiting and programmatic enforcement of robots.txt protocols to ensure continuous, compliant data availability.
- Developed automated sanitization scripts to purge unstructured “noise” (such as emojis and broken URLs) from thousands of raw posts, transforming messy social streams into a structured dataset viable for analysis.
- Architected an NLP preprocessing workflow using spaCy and IBM Watson to extract “Big Five” personality traits, reducing dataset noise by ~45% and improving downstream model accuracy by ~30%.

Software Engineer (Revenue Operations), Free Pixel Games Limited, Bangladesh [November 2020 – May 2021]

Unified Ad-Service Revenue Dashboard

- Solved the “N+1 format” problem by designing a normalized MySQL data warehouse that standardized conflicting revenue data from 5+ disparate ad networks.
- Automated the daily revenue reconciliation workflow, reducing manual data consolidation time by >80% (from 2 hours to 15 mins).
- Developed a predictive anomaly detection engine that triggered real-time alerts if Revenue-Per-Mille (eCPM) dropped by >10%, preventing revenue leakage.

Additional Experience:

Microsoft Senior Advisor, Best Buy Canada Ltd. Halifax, NS [October 2023 – Present]

- Standardized the sales training methodology for Microsoft 365 (Excel/Teams) products, ensuring the team adhered to consistent operational procedures.
- Outperformed store-wide sales efficiency benchmarks by ~35% (revenue-per-hour) by utilizing a consultative approach to translate complex client requirements into tailored IoT and computing architectures.
- Contributed approx. 29% of total departmental revenue through precise technical needs assessment and solution mapping, ensuring hardware compatibility and budget alignment.

Education:

- **Master of Applied Computer Science** [September 2023 – May 2025]
Dalhousie University, Halifax, NS
- **Master’s in Applied Statistics and Data Science** [November 2019 – April 2022]
Jahangirnagar University, Bangladesh
- **BSc. in Computer Science and Software Engineering** [January 2015 – January 2020]
American International University-Bangladesh

Core competencies:

- Data Quality & Governance: Python (Pandas/NumPy), SQL (PostgreSQL/MySQL), ETL Pipelines, Data Cleansing, Root Cause Analysis, Automated Validation Scripts.
- Reporting & Visualization: Tableau, Looker Studio, Dash/Plotly, Advanced Excel (Pivot Tables, VLOOKUP), JIRA, Confluence.
- Infrastructure & Tools: AWS (Lambda, S3), Docker, Git, REST APIs, MS Office 365.